

AI Agents

Session 1 — Fundamentals, landscape, and your first agent

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Master's in Multivariate Statistics · USAL

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Who we are

Me

- Who am I?
- Why I teach this class
- How I work day-to-day with agents

Who we are

You

- Physicists, mathematicians, statisticians
- Engineers, computer scientists
- Biologists, psychologists
- Intermediate technical level
- **Your advantage:** you already think in terms of uncertainty, evidence, variance

Initial poll

Raise your hand:

1. Who has used ChatGPT / Claude / Gemini this week?
2. Who has ever called an LLM API from code (Python/R)?
3. Who has heard of MCP, function calling, or tool use?
4. What do you hope to take away? (one word)

| The goal is to calibrate the pace, not to judge.

Agenda — Session 1

Block	Content
0	Welcome + poll
1	Fundamentals: what is an agent, brain, hands, memory, skills
	Break
2	Landscape of frameworks and existing agents
3	Agents you can already use today
4	"No-code" workshop: your first local agent
5	Wrap-up and discussion

Logistics

-  **Laptops open:** you'll be installing things
-  **Materials:** this deck + other resources available at <https://github.com/cbjuan/2026-agents-usal>
-  **Questions:** whenever you want, no need to wait until the end
-  **Technical issues during the workshop:** we solve them as a group

| If something doesn't work on your laptop during the workshop, **it's perfectly normal**. It's part of the learning.

Materials



github.com/cbjuan/2026-agents-usal

Scan the QR or type the URL. Inside you'll find:

- Both decks (source `.md` + PDF)
- Workshop code templates
- Datasets used in the tasks
- Reading list with links
- Troubleshooting notes

Shared vocabulary

Terms you'll hear repeatedly — this slide stays in the deck for reference:

Term	Meaning
LLM	Large Language Model — GPT, Claude, Gemini, Qwen...
API	Application Programming Interface — calling a service from code
Token	Sub-word unit the LLM reads/writes ($\sim\frac{3}{4}$ of a word in English)
Prompt	Text input to the LLM; system prompt = fixed instructions
Context window	Max tokens the LLM can see at once (4K, 32K, 200K...)
Tool-call	LLM invoking a function (search, read_file, run_sql...)
Inference	One forward pass through the model = cost + latency
RAG	Retrieval-Augmented Generation — fetch docs, add to prompt
MCP	Model Context Protocol — standard to connect tools to LLMs
Fine-tuning	Extra training on specific data to adjust a model's behavior

BLOCK 1

Fundamentals

What (and what isn't) an agent?

The word is everywhere. It means different things:

- In classical AI: an object that **perceives** and **acts upon** an environment
- In products: any app with an LLM
- In venture capital: whatever can raise a Series A

We're not going to invent a canonical definition.

We'll use **five complementary lenses** and pick the useful one for each context.

5 lenses for looking at an agent

Lens	Where it comes from	What it contributes
AIMA (Russell & Norvig)	Classical AI · 1995→	Taxonomy + connection to MDPs/RL
Anthropic	Building Effective Agents · 2024	<i>Workflow vs agent</i>
OpenAI	Practical guide to building agents · 2025	Model + Tools + Instructions
LangChain (H. Chase)	Cognitive architectures	Autonomy levels 1→6
CoALA (Princeton)	TMLR 2024	Memory + actions + decision

And Simon Willison's operational rule: *"an LLM agent runs tools in a loop to achieve a goal."*

Lens 1 · AIMA (Russell & Norvig)

"Anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators."

— *Artificial Intelligence: A Modern Approach* (1995→2020)

What matters: the definition does not require language, intelligence, or LLMs.

A thermostat is an agent. A human is an agent. A program too.

What does define an agent: having an explicit or implicit **utility function**, and choosing actions that maximize it.

AIMA taxonomy · 5 levels

1. **Simple reflex** — *condition* → *action* rule on the current percept
2. **Model-based reflex** — maintains internal state (partially observable environments)
3. **Goal-based** — picks actions that bring it closer to a goal
4. **Utility-based** — compares states with a utility function
5. **Learning** — adds *learning element* + *performance element* + *critic*

An LLM with tools and a loop ≈ hybrid between goal-based and learning-by-context.

Connection with MDPs and RL (the statistician's lens)

An agent can be read as a *policy* over a **Markov Decision Process** (MDP) $\langle S, A, P, R, \gamma \rangle$:

S = states · A = actions · $P(s' \mid s, a)$ = transition probabilities · $R(s, a)$ = reward · $\gamma \in [0, 1]$ = discount factor

- s_t = LLM context (messages + observations)
- a_t = token-stream that produces a *tool-call* or response
- r_t = test that passes, human reward, automated eval
- π_θ = the LLM with its prompt and tools

Three classical **reinforcement learning** (RL) problems reappear identically:

- **Credit assignment** in multi-turn (which step failed?)
- **Exploration** (call a new tool?)
- **Long horizon** — errors compound

Lens 2 · Anthropic — *workflow vs agent*

Workflow

Predefined steps in code.

The programmer owns control flow.

```
input → classify → template  
      → policy → send
```

Predictable. Auditable. Cheap.

Agent

LLM **decides** which tool to call and when to stop,
in a loop.

The model owns control flow.

```
loop: LLM → tool → observation  
      → stop?
```

Flexible. Expensive. Non-deterministic.

Heuristic: if you can map the decision tree, **don't build an agent.**

Lens 3 · OpenAI

"A system that independently accomplishes tasks on your behalf."

— *A practical guide to building agents*, OpenAI · 2025

Three components:

- **Model** — the brain that reasons
- **Tools** — *data, action, orchestration*
- **Instructions** — policy, format, *guardrails* (safety filters and limits)

Minimal pattern: `while` loop with an exit condition.

When to split into multi-agent: complex logic or many overlapping tools.

Lens 4 · LangChain — autonomy levels (synthesis)

Level	Who decides	Example
1. Code only	The human (program)	<code>df.describe()</code>
2. LLM call	Human + LLM once	"summarize this text"
3. Chain	Program with several LLMs in fixed order	Classic RAG
4. Router	LLM picks a branch	classifier → specialist
5. State machine	LLM inside a graph with transitions	LangGraph
6. Autonomous agent	LLM in a free loop	Claude Code, Cursor

Harrison Chase (LangChain CEO): *"outsource your agentic infrastructure, but **own** your cognitive architecture."*

Lens 5 · CoALA (Cognitive Architectures for Language Agents)

Sumers, Yao, Narasimhan, Griffiths · Princeton · TMLR 2024

An LLM agent has three axes:

Memory (LTM = long-term memory)

- *Working* (in-context)
- Episodic LTM
- Semantic LTM
- Procedural LTM

Actions

- Internal (on memory itself)
- External (on the environment)

Decision

- Planning
- Execution
- Reflection

ReAct, Reflexion, Voyager, Generative Agents... they all fit here.

"An LLM agent runs tools in a loop to achieve a goal."

— [Simon Willison](#) · September 2025 (after collecting >200 distinct definitions)

When NOT to build an agent

Three questions (Barry Zhang, Anthropic):

1. **Is the task ambiguous?** If you can map the entire decision tree → **workflow**
2. **Is it worth the cost?** If it's a small, cheap task, an agent is overkill
3. **Is the model capable of the atomic task?** Errors compound multiplicatively

"It is the decade of agents, not the year of agents."

— Andrej Karpathy · [Dwarkesh Podcast, Oct 2025](#)

Errors accumulate multiplicatively. Best to assume it will take time.

Spectrum: chatbot → RAG → workflow → agent

Pure chatbot

- └ One LLM call, no state, no tools

RAG

- └ LLM + pre-inference retrieval · fixed structure

Workflow

- └ LLM(s) + tools on code-predefined routes

Agent

- └ LLM + tools in a loop, decides flow, its own stopping criterion

Most "agentic applications" are workflows with an LLM inside.

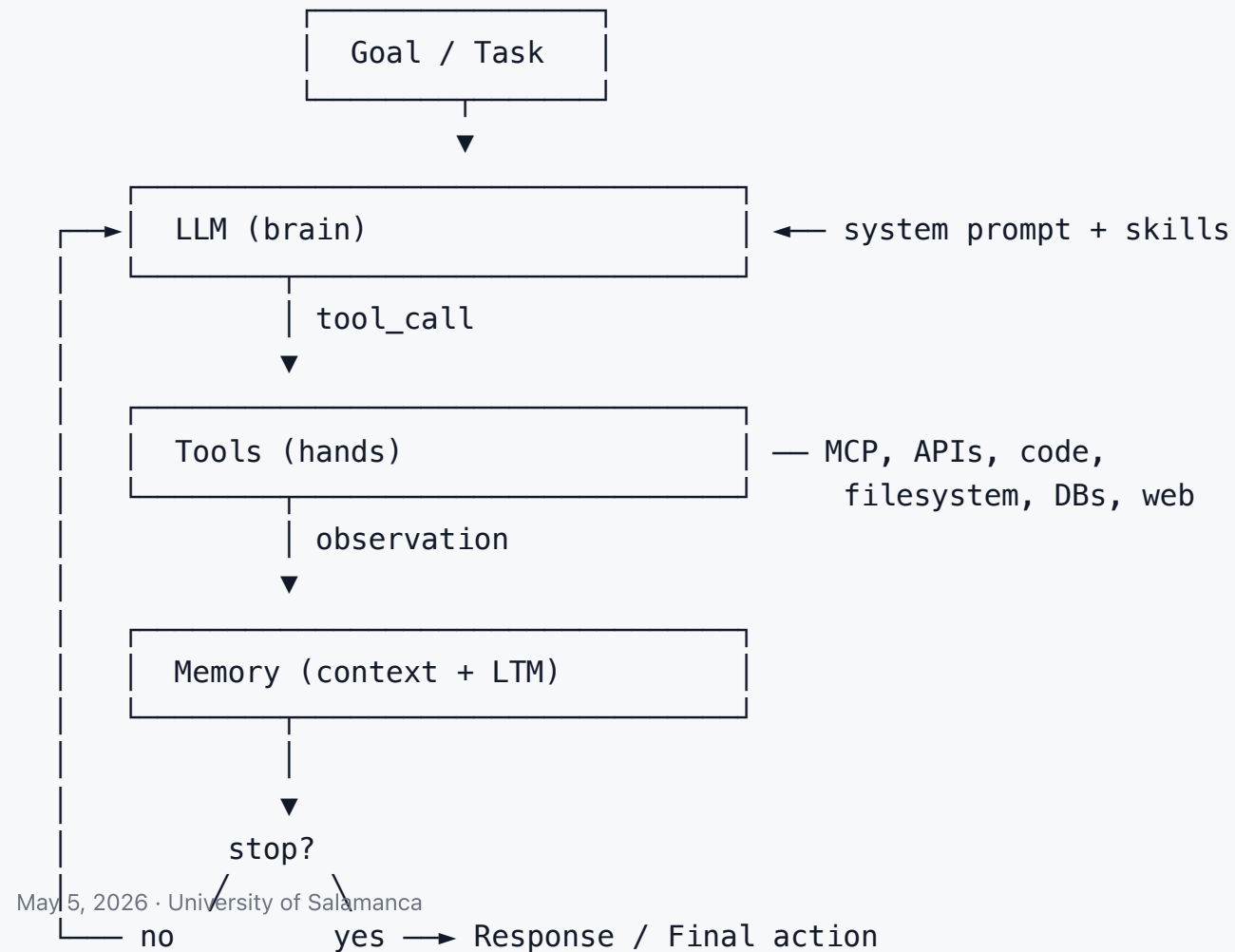
Discussion question

To detect **fraud in bank transactions in real time**, would you use an agent or a workflow?

Why?

Discuss with the person next to you. Then we share.

Typical architecture · the agentic loop



The brain · reasoning models in 2026

Frontier — top-tier closed models, accessed via API

- **Claude** (Anthropic) — Opus 4.x, Sonnet 4.x
- **GPT-5.x** (OpenAI)
- **Gemini 3.x** (Google)
- **Grok 4** (xAI)

Open-weights — weights publicly downloadable, runnable on your laptop or server

- **Qwen3 / Qwen3.5 / Qwen3.6** (Alibaba) — Apache 2.0
- **Granite 4.1** (IBM) — Apache 2.0
- **DeepSeek V4 / R1** — large **MoE** (mixture-of-experts) models
- **Phi-4 / Phi-4-mini** (Microsoft) — MIT
- **Gemma 3 / 4** (Google), **SmolLM3** (HuggingFace), **Mistral 3**

Frontier vs open-weights — when which?

Criterion	Frontier (API)	Open-weights (local)
Reasoning quality	★★★★★	★★★★ – ★★★★★
Complex tool-use	Very reliable	Reliable on Qwen3+, Granite 4.1
Cost per task	\$\$	\$ (electricity)
Latency	Cloud	Depends on hardware
Privacy	Data to the provider	100% local
Reproducibility	Changes without notice	Pinnable versions
Fine-tuning	Limited	Full control

In the workshop we'll use open-weights, not out of ideology, but so it works without an account and without spending money.

Lightweight open-weights · top 3 for the workshop

Model	RAM Q4	Tool-use	License	Command
Qwen 3.5 4B	~3.5 GB	Excellent (native)	Apache 2.0	<code>ollama pull qwen3.5:4b</code>
Qwen 3.5 9B ★	~6.5 GB	Excellent (native)	Apache 2.0	<code>ollama pull qwen3.5:9b</code>
Granite 4.1 8B	~5.3 GB	Solid on BFCL	Apache 2.0	<code>ollama pull granite4.1:8b</code>
Gemma 4 e2b	~2.5 GB	Native function-calling	Apache 2.0	<code>ollama pull gemma4:e2b</code>
Gemma 4 e4b	~4.5 GB	Native function-calling	Apache 2.0	<code>ollama pull gemma4:e4b</code>

Q4 = 4-bit quantization (model compressed to $\frac{1}{4}$ the memory, small quality loss).

BFCL = Berkeley Function Calling Leaderboard, a tool-use benchmark.

Why Qwen 3.5 (not 3.6): Qwen 3.6 only ships at 27B+ — too large for laptops.

For 32 GB, **Gemma 4 26B** is a fast MoE option: `ollama pull gemma4:26b` (~18 GB).

Also solid: **Granite 4.1 30B** (~17 GB) and **Nemotron Cascade-2 30B MoE**.

The hands · function calling

How it works in 3 steps:

```
# 1. Describe the tool as a JSON schema
tools = [{
    "name": "descriptive_stats",
    "description": "Compute descriptive statistics of a CSV",
    "parameters": {
        "type": "object",
        "properties": {"path": {"type": "string"}},
        "required": ["path"]
    }
}]

# 2. The LLM responds with a structured object, not text
# {"tool": "descriptive_stats", "args": {"path": "iris.csv"}}

# 3. Your runtime executes the tool and returns its output to the LLM
```

Important: the "magic" lies in the model's *fine-tuning* to emit valid JSON when tools are in the prompt.

MCP · Model Context Protocol

What it is: a universal **JSON-RPC** (remote procedure calls encoded in JSON) protocol between **clients** (Claude Desktop, Cursor, ChatGPT) and **servers** (processes that expose tools, resources, prompts).

Origin: Anthropic, Nov 25, 2024. **Open source** (Linux Foundation since Dec 2025).

Why it matters:

- Moves from **N×M** ad-hoc integrations to **N+M**
- Once you have an MCP server for your DB, **any client** can use it
- De facto standard in 2026

Five primitives: tools, resources, prompts, sampling, roots

Two transports: stdio (local) and Streamable HTTP (remote, OAuth 2.1)

MCP · 2026 adoption state

- **Thousands of MCP servers** published (public registry + private)
- **Tens of millions of monthly downloads** of the SDK (software development kit — the library devs use)
- **Most enterprise teams** report at least one MCP-based agent in production

Native support in:

Claude · ChatGPT (Apps SDK) · Gemini API · Vertex AI · Cursor · Windsurf · Zed · JetBrains AI · OpenAI Agents SDK · Vercel AI SDK

Adjacent ecosystem:

- **A2A** (Google, under Linux Foundation) — agent ↔ agent, JSON-RPC
- **ACP** (IBM / BeeAI, now merged into A2A) — REST-native agent interop
- **AG-UI** — frontend ↔ agent

Ttools · live example · setup

Instructor demo — same stack you'll install in Block 4: OpenCode + Ollama + a local open-weights model.

Launch OpenCode — model comes from the config (`ollama/gemma4`):

```
ollama launch opencode --model gemma4
```

Tools · live example · run

Ask in the chat:

| *"List the CSVs in my Downloads folder"*

Under the hood:

1. OpenCode discovers the tools
2. Gemma emits a structured call to the proper tool
3. OpenCode runs it locally → returns the file listing
4. Gemma loops (read each CSV → count rows) → composes the final answer

No cloud. No API key. Same protocols as Claude Desktop and Cursor use.

Memory · four types

Type	What it stores	Examples
Working / context	Active tokens	The LLM itself
Episodic	"What I did last week"	Mem0, Letta, Zep, MemMachine
Semantic	Facts, profiles, KG	Qdrant, pgvector, Neo4j
Procedural	Rules, <i>skills</i> , scripts	Claude Skills, AGENTS.md

CoALA framework: human memory serves as inspiration, not as an exact map.

Context engineering > prompt engineering

| *"The set of strategies for curating and maintaining the optimal set of tokens during inference."*

Mindset shift:

-  From: picking the best 5 sentences of the system prompt
-  To: designing **what information arrives when**

Key techniques:

- **Just-in-time retrieval** — `glob` and `grep` explore on demand
- **Tool-mediated retrieval** — the LLM asks for what it needs
- **Compression** — summary between steps
- **Pruning** — discard the irrelevant
- **Hierarchical loading** — *progressive disclosure* (load detail only when needed)

Skills · *progressive disclosure*

```
---  
name: pdf-processing  
description: Extract text and tables from PDF files, fill forms.  
            Use when the user mentions PDFs or document extraction.  
---  
  
# Instructions  
1. ...
```

Loads in three levels:

1. **Frontmatter** always in context (~50 tokens)
2. **Body** loads if the skill activates
3. **Sub-files** only if relevant

Decouples the **size** of the skill from its **cost per token**.

Skills vs MCP vs subagents

Concept	What it provides	When to use it
MCP	Access to tools / data	Connecting to external systems
Skill	Procedure / knowledge	Teaching how to do something
Subagent	Process with its own context	Isolate context, parallelize

They're complementary, not mutually exclusive.

Real example: a "statistical analysis" *skill* uses the internal DB's *MCP* and delegates to a "QA" *subagent* to review the output.



BLOCK 2

Framework landscape

and existing agents

Open source frameworks · 2026 landscape

Framework	Philosophy	Curve
LangGraph	Directed graph with state, checkpoints, HITL	High
CrewAI	Roles + goals + tasks ("crew")	Low
AutoGen / AG2	Multi-agent conversations	Medium
OpenAI Agents SDK	Handoffs between agents	Low
Google ADK	Hierarchical tree, native A2A	Medium
PydanticAI	Extreme type-safety	Medium
Smolagents (HF) ★	~1000 LOC, <i>CodeAgent</i> , sandbox	Very low
Agno	Multi-agent runtime with good DX	Medium
LlamaIndex Agents	Centered on RAG over data	Medium
DSPy	Optimizable prompt programming	High (different paradigm)

HITL = human-in-the-loop · DX = developer experience · LOC = lines of code · SDK = software development kit.

How to choose a framework (simple matrix)

	Single	← Composition →	Multi-agent
Low flow control	smolagents PydanticAI	· ·	CrewAI AutoGen
High flow control	LangChain classic	· ·	LangGraph Microsoft AF

Harrison Chase's rule of thumb:

*"Outsource your agentic infrastructure, but **own** your cognitive architecture."*

Coding agents · the most mature on the market

Why coding first: tests are verifiable → clear reward signal.

- **Claude Code** (Anthropic, terminal) — current benchmark
- **Cursor / Windsurf** (IDEs)
- **GitHub Copilot Coding Agent**
- **OpenAI Codex** (with Background Computer Use)
- **Devin** (Cognition) — expensive, variable performance depending on benchmark
- **Replit Agent** — long sessions, autonomous
- **OpenCode / Pi** — open source, terminal

| It's the area where agents work best today.

Vibe coding and app builders

They build full-stack apps from natural language:

- **Lovable · Bolt · v0 (Vercel) · Same.dev**
- **Manus · Leap.new · Replit**

Useful for: quick prototypes, landing pages, demos.

Not useful for: real production, refactors, complex systems.

"The fastest way to build software is to vibe code your idea, then have someone rewrite it properly." — the line you hear in SF in 2026.

Computer use · agents that move the mouse

- **Anthropic Computer Use** — receives a screenshot, returns actions
- **ChatGPT Agent Mode** (OpenAI, integrates the old Operator)
- **Google Gemini Computer Use** / Project Mariner
- **Manus AI** — "general VM (virtual machine) agent"

Real state: they work, but they're **slow** (seconds per action), **fragile** (UI changes break them), **expensive**.

OSWorld benchmark: performance still clearly below expert human.

Browser agents · the next frontier

- **Perplexity Comet** (free since Oct 2025)
- **ChatGPT Atlas** (Oct 2025, macOS only)
- **Claude for Chrome** (extension)
- **Edge Copilot Mode · Opera Aria/Neon · Brave Leo**
- **BrowserOS** (open source, Chromium fork with Ollama)

Advantages: session, cookies, login already there.

Risks: prompt injection from any visited page (we'll cover this tomorrow in security).

Open legal case · Amazon vs Perplexity

January 2026: Amazon sues Perplexity for using Comet agents to automate purchases, alleging a bypass of Amazon's **Terms of Service (TOS)**.

Perplexity's position: the agent acts as a human, identifying itself with a standard User-Agent.

Amazon's position: automation at scale violates the terms of service.

Question for the group:

Is this legitimate TOS defense or resistance to change?

Who bears responsibility when an agent "clicks"?

BLOCK 3

Agents you can already use today

Recommended stack for a statistics / data profile

Analysis and code

- **Claude Code, OpenAI Codex or Cursor** for scripts
- **Code Interpreter / Artifacts** for ad-hoc analysis
- **OpenCode + Ollama** local (today's workshop)

Research and synthesis

- **Perplexity / Perplexity Pro Search**
- **NotebookLM** for synthesizing papers
- **Deep Research** (Anthropic, OpenAI, Gemini) or **Elicit** for systematic review

| There's no "correct" stack. What kills you is not trying.

Quick demos

1. **Perplexity Pro Search** — *"robust methods in PCA, 2024-2026 papers"*
2. **NotebookLM** — 3 papers in Spanish → mind map + podcast
3. **Claude / ChatGPT** with artifacts — live CSV analysis
4. **Claude Computer Use** — open Excel, fill out a summary table

What you'll see: the difference between a *chatbot* (saying things) and an *agent* (doing things).

BLOCK 4

Workshop: your first agent without code

Workshop plan

Min	Activity
0–10	Instructor demo: Claude Code in action
10–15	Install Ollama (in parallel)
15–20	Download the recommended model
20–25	Install + configure OpenCode
25–55	Open task: agent analyzes a CSV and generates a report
55–60	We share results

Prerequisites: ≥ 8 GB RAM, ≥ 15 GB free disk, install permissions.

Instructor demo · Claude Code

Example task:

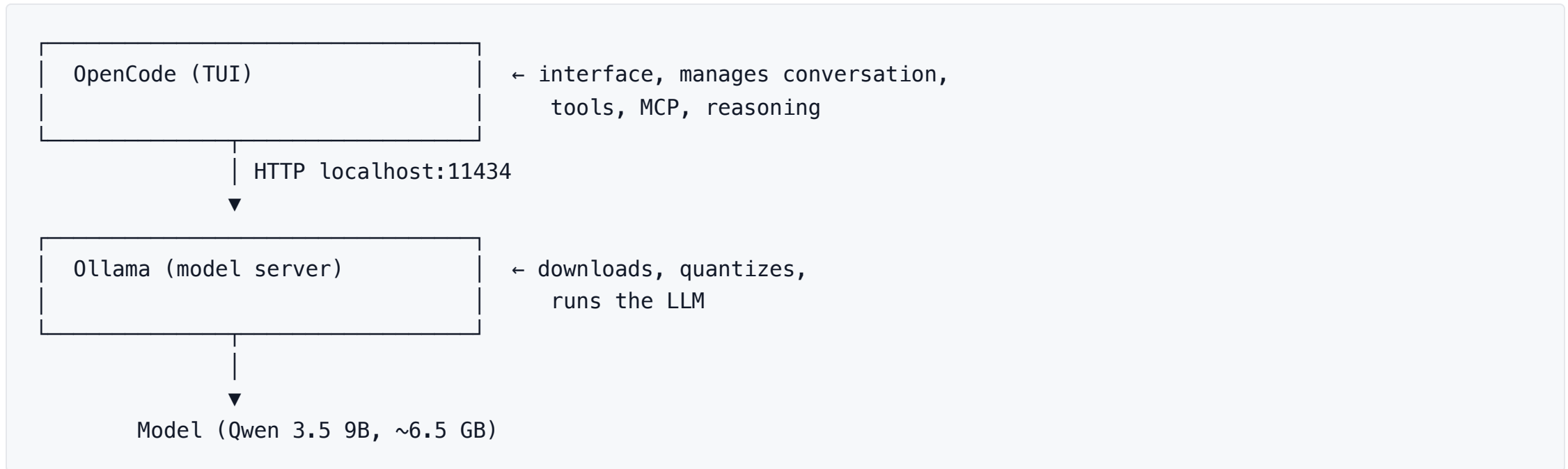
| *"Add NumPy-style docstrings to this module and generate a README based on the tests."*

Show:

- Subagents (Explore, Plan)
- Dynamically loaded skills
- Connected MCP servers (filesystem, git)
- Visible reasoning trace
- When the agent asks for confirmation (**human-in-the-loop / HITL**)

| This is the "ceiling" of what's possible today. You'll build something simpler, but of the same lineage.

Your stack · Ollama + OpenCode



TUI = terminal user interface (text-only, no mouse). Quantize = compress a model to fewer bits per weight.

Both open source. Run offline. No account. Cross-platform.

Step 1 · Install Ollama

```
# macOS / Linux
curl -fsSL https://ollama.com/install.sh | sh

# Windows: download installer from ollama.com

# Verify
ollama --version
```

Check that the server responds:

```
curl http://localhost:11434/api/tags
```

If it returns JSON, you're good.

Step 2 · Download the model

```
# 8 GB RAM
ollama pull qwen3.5:4b      # ~3.5 GB

# 16 GB RAM (recommended)
ollama pull gemma4:e4b      # ~9.4 GB
ollama pull qwen3.5:9b      # ~6.5 GB

# 32 GB RAM (fast MoE)
ollama pull gemma4:26b      # ~18 GB
```

Smoke test:

```
ollama run gemma4 "Hi, what model are you?"
```

⚠ If tools fail later in the workshop, suspect the default `num_ctx` (4K). You need to raise it to 32K.

Step 3 · Install OpenCode

```
# Recommended
curl -fsSL https://opencode.ai/install | bash

# Alternatives
npm install -g opencode-ai@latest
brew install sst/tap/opencode

# Verify
opencode --version
```

Create a config file at

`~/.config/opencode/opencode.json` (Linux/macOS)
or `%APPDATA%\opencode\opencode.json` (Windows).

Step 3b · Minimal `opencode.json`

```
{
  "$schema": "https://opencode.ai/config.json",
  "provider": {
    "ollama": {
      "models": {
        "gemma4": {
          "name": "gemma4",
          "tools": true,
          "reasoning": true,
          "options": { "num_ctx": 32768 }
        },
        "qwen3.5:4b": {
          "name": "qwen3.5:4b",
          "tools": true,
          "reasoning": true,
          "options": { "num_ctx": 32768 }
        }
      }
    },
    "name": "Ollama",
    "npm": "@ai-sdk/openai-compatible",
    "options": {
      "baseUrl": "http://127.0.0.1:11434/v1"
    }
  }
}
```

Step 4 · Workshop tasks

Each student picks ONE. Work in pairs if you like.

1. Iris analyst
"Download the Iris dataset from sklearn, compute the correlation matrix and save it as heatmap iris_corr.png."
2. Doc generator
"Generate a Markdown that explains what an agent is, citing 3 web sources. Save it as agent_intro.md."
3. CSV explorer
"Read sales.csv (I'll provide it) and compute mean, median, and standard deviation by category."

Launch OpenCode:

```
mkdir ~/2026-agents-usal && cd ~/2026-agents-usal  
opencode
```


Troubleshooting · if something fails

Symptom	Likely cause	Fix
Model replies with text but doesn't call tools	<code>num_ctx</code> too low	Raise to 16K-32K in <code>opencode.json</code>
Connection error	Ollama not running	<code>ollama serve</code> in another terminal
Streaming gets cut	Undici timeouts	Update OpenCode to the latest release
Out of memory	Model too large	Switch to <code>qwen3.5:4b</code> or <code>gemma4:e2b</code>
Tool-call fails with many tools	Hermes-style hallucinates	Reduce tools, raise <code>num_ctx</code>
MCP server SSE (Server-Sent Events) gives <code>UnknownError</code>	Known bug	Use streamable HTTP transport

Pi — minimalist alternative (brief mention)

`pi-mono` (by Mario Zechner / *badlogic*) is a deliberately **anti-feature** *coding agent*:

- **No** MCP, no sub-agents, no permission popups
- Just 4 core tools (read/write/edit/bash)
- Philosophy: *"build CLI tools with READMEs, the agent reads them on demand"*

Useful as a **pedagogical provocation**: where should complexity live? In the harness or in traditional Unix tools?

Don't install it today. Mentioned for the curious.

Hermes Agent — self-evolving alternative (brief mention)

`hermes-agent` (Nous Research) takes the **opposite bet** to pi:

- **Autonomous skill creation** — generates and refines procedures from experience
- **Persistent memory** across sessions, full-text session search
- **Multi-platform gateway** — Telegram, Discord, Slack, WhatsApp, Signal from one backend
- Runs on anything from a \$5 VPS to GPU clusters · MIT licence
- Companion repo `hermes-agent-self-evolution` auto-optimises skills + prompts via DSPy + GEPA

Pedagogical provocation: where should the agent's knowledge live? Baked into Unix tools (pi) or evolved at runtime (hermes)?

Don't install it today. Mentioned for the curious.

OpenClaw — batteries-included, local-first (brief mention)

[OpenClaw](#) (formerly ClawdBot) sits at the opposite end from pi: *everything bundled, everything local*.

- **Multi-platform** — WhatsApp, Telegram, Discord, Slack, Signal, iMessage
- **50+ integrations** out of the box (Gmail, Notion, HomeKit, browser automation...)
- **Privacy-first** — data stays on your machine unless you explicitly send it out
- **Model-agnostic** — Claude / GPT / Gemini / Llama / Mistral + **Ollama** + LM Studio
- **MIT, TypeScript** · `npm i -g openclaw` · repo [openclaw/openclaw](#)

One axis, three takes: **pi** (nothing included) → **Hermes** (self-evolving) → **OpenClaw** (batteries-included, local).

Don't install it today. Mentioned for the curious.

Wrap-up

What we've seen today

- ✓ Five lenses to define an agent · when NOT to use one
- ✓ Loop: brain + hands + memory + control
- ✓ 2026 models · frontier vs open-weights
- ✓ Function calling · MCP · Skills
- ✓ Frameworks · vertical agents · computer use · browser agents
- ✓ Your first local agent up and running

Tomorrow: how they're built, how they're evaluated, how they break.

Reading for tonight (optional, ~30 min)

Essential:

- Anthropic — [*Building Effective Agents*](#) (Dec 2024)
- Anthropic — [*Effective Context Engineering*](#) (2025)
- HuggingFace — [*Introducing smolagents*](#) (2024)

Supplementary:

- Karpathy on Dwarkesh — [*AGI is still a decade away*](#) (Oct 2025)
- Kambhampati et al. — [*LLMs Can't Plan, But Can Help Planning*](#) (ICML 2024)
- Simon Willison — [*agent-definitions*](#) [tag](#)

| If you read only one: Anthropic's [***Building Effective Agents***](#).

Question for tomorrow

*Which of your tasks would clearly benefit from an **agent** vs. a **rigid workflow**?*

Think it over tonight. We'll discuss at the start tomorrow.

See you tomorrow! 🙌

May 6 · 16:00

Building, evaluating, and securing agents

Thank you.